Modeling the Spread of Epidemics

SIS, SIR, and Stochastic Network Perspectives

Ayush Mahale

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Abstract

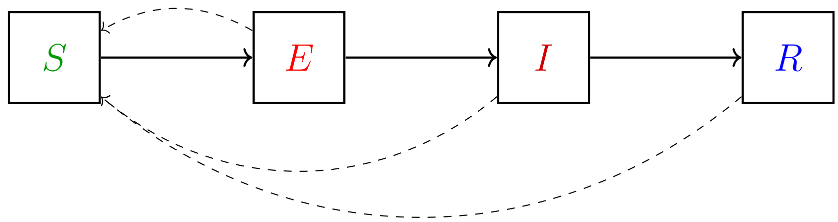
This paper studies the spread of epidemics through the use of mathematical models, focusing on compartmental models such as SIS and SIR, along with stochastic formulations on networks. Using differential equations, we analyze short and long term behavior of infection dynamics, including threshold phenomena governed by the basic reproduction number. We also take a closer look at SIR models and their connection to branching processes and infection digraphs, which provide a probabilistic perspective on the spread of epidemics in large populations.

1 Modeling Epidemics

We can model the spread of epidemics with the following possible states (where arrows represent the main progression, and dashed arrows represent alternative paths):

- **S** – Susceptible
- **E** – Exposed
- **I** – Infectious
- **R** – Recovered

This is an example of a Compartment model, where we divide the population into groups and model how people move between them. We will go through the following 3 approaches to analyze the Compartment model:



- Differential Equations
- Stochastic Models
- Branching Process

In this scribe, we will primarily analyze the Compartment model, but there are 2 other ways to model the spread of an epidemic through a network:

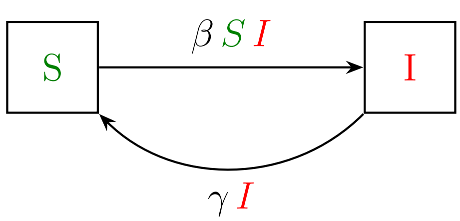
1.) Spread on Networks: This approach involves modeling individuals as nodes in a graph, connected by edges (where an infection can be transmitted via an edge).

2.) Agent-Based Models: This approach involves governing the behavior of individual agents to simulate the propagation of infections in a population.

2 Differential Equations Approach

2.1 SIS Model

Let's take a closer look at a simplified SIS model, where individuals can move from Susceptible to Infectious, and vice versa (so individuals cannot become immune to the infection, and must leave the current state eventually).



Where β is the infection rate, and γ is the recovery rate. Therefore, βSI is the rate at which we move from susceptible to infectious, and γI is the rate at which we move from infectious to susceptible.

We know that $S+I=N$, let's replace N by 1 and work with percentages (so 100 percent of the population lies in either Infected or Susceptible). Therefore, $S+I=1$. We can model the change in rate of infections and recoveries by considering the net flux of mass as follows:

$$\frac{dI}{dt} = \beta(1-I)\,I - \gamma I \tag{1}$$

$$\frac{dS}{dt} = -\beta SI + \gamma I \tag{2}$$

2.2 Short Time Behavior of SIS Model

Let's consider the behavior of the SIS in the short time horizon, specifically when t is near 0. We consider the initial conditions

$$I(0) = I_0 > 0$$

and define *basic reproduction number*

$$R_0 = \frac{\beta}{\gamma}$$

where:

- $I(0)$ represents the initial proportion of individuals infected.
- R_0 represents the expected number of new infections caused by a single person, so we expect:
  - if $R_0 > 1$ the infection spreads into an outbreak and the number of infected people spreads exponentially for small t.
  - if $R_0 < 1$ the infection dies out following an exponential decay.

To analyze how fast the infection grows or decays, let's consider the log-derivative of infected individuals:

$$\frac{d}{dt}\log I = \frac{1}{I}\frac{dI}{dt} = \beta(1-I) - \gamma = \gamma\big[R_0(1-I) - 1\big] \tag{3}$$

If the log-derivative is positive, the infection grows exponentially, and if the log-derivative is negative, the infection decays exponentially.

Case 1: $R_0 = \frac{\beta}{\gamma} < 1$, implying recovery happens faster than infection since $\gamma > \beta$.

$$\frac{d}{dt}\log I = \gamma\big[R_0(1-I) - 1\big] \;\leq\; R_0 - 1 \tag{4}$$

showing that

$$I(t) \;\le\; I_0\, e^{-\gamma(1-R_0)\,t} \tag{5}$$

Case 2: $R_0 = \frac{\beta}{\gamma} > 1$, implying that initially, recovery happens slower than infection since $\gamma < \beta$.

Indeed, for I_0 sufficiently small, we have $R_0(1 - I_0) > 1$ showing that initially, $\frac{d}{dt} \log I > 0$ so that for small times $I(t)$ grows exponentially,

$$I(t) \approx I_0\, e^{\gamma\big(R_0(1-I_0)-1\big)\,t} \tag{6}$$

2.3 Large time behavior

Next we analyze the large time behavior of an SIS epidemic, in particular the stationary infection rate $I(\infty)$. We will look at both of the possible cases, $R_0 \le 1$ and $R_0 > 1$.

Case 1: $R_0 \le 1$**.**

Under this condition, $\beta \le \gamma$, so by (1),

$$\begin{aligned} \frac{dI}{dt} &= \beta(1-I)\,I - \gamma I \\ &= \big[(\beta-\gamma) - \beta I\big]\, I \\ &\le -\beta I^2 \end{aligned} \tag{7}$$

Thus as long as $I(t) > 0$, $\frac{dI}{dt} < 0$, showing that I is monotonically decreasing and approaches 0 as $t \to \infty$. Note also that the above equation implies that $dI/dt > -\gamma I$, so I can at best fall exponentially with rate γ, which means for all finite times it will be strictly > 0 if it starts off with $I_0 >$.

Since the sum of the susceptible fraction and infectious fraction entails the entire population $S + I = 1$:

$$S(t) = 1 - I(t) \nearrow 1, \tag{8}$$

unless $I(0) = 0$, in which case $I(t) = 0$ and $S(t) = 1$ for all t, see Fig. 1 below.

Case 2: $R_0 > 0$**.**

Consider the log-derivative in equation (3):

$$\frac{d}{dt} \log I = \gamma[R_0(1-I) - 1]$$

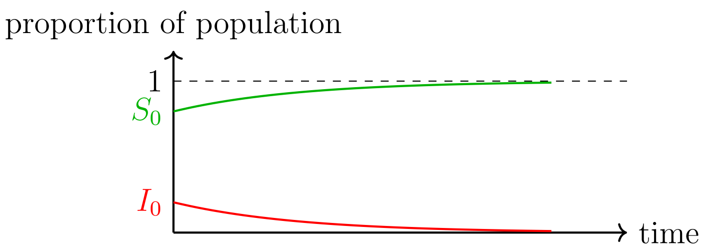


Figure 1: Visualization of Case 1, $R_0 \leq 1$, Susceptible proportion tends to 1 and Infectious proportion tends to 0.

Defining I_∞ by setting the right hand side to zero,

$$I_\infty = 1 - \frac{1}{R_0}.$$

we can rewrite this equation as

$$\frac{d}{dt} \log I = \gamma R_0 (I_\infty - I) \tag{9}$$

This shows that $I(t)$ is

- increasing when $I(t) < I_\infty$,
- decreasing when $I(t) > I_\infty$,
- constant when $I(t) = I_\infty$.

In particular, I_∞ is the unique stationary point where I does not change. To discuss the general case, starting from an arbitrary I_0, we denote the right hand side of (9) by $f(I)$, and then distinguish the two cases $I_0 < I_\infty$ and $I_0 > I_\infty$.

- If $I_0 < I_\infty$, then $f(I_0) > 0$, $\dfrac{d}{dt} \log I > 0$ and $I(t)$ grows. As I grows, $f(I)$ moves downward (since we have a negative slope), so the growth rate monotonically decreases until it reaches $f(I) = 0$, and I asymptotically approaches I_∞ from below as seen in Figure (2).
- If $I_0 > I_\infty$, then $f(I_0) < 0$, $\dfrac{d}{dt} \log I < 0$ and $I(t)$ decays. While I falls, $f(I)$ slides upward, slowing the decay until again $f(I) = 0$ and I approaches I_∞ from above this time, see Figure (3).

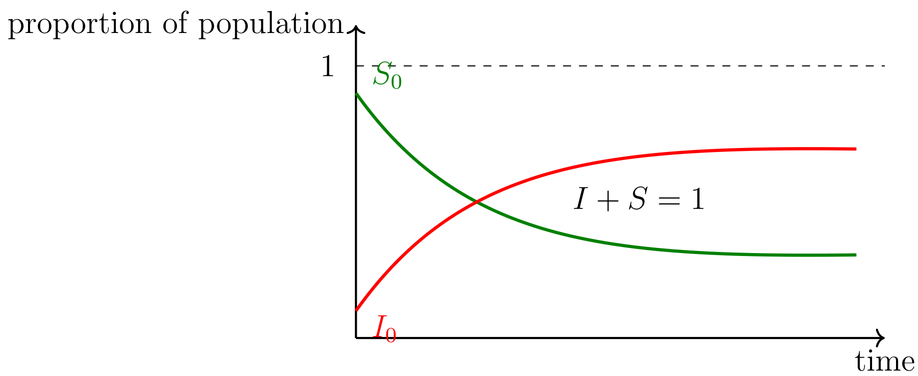


Figure 2: Visualization of Case 2 $R_0 > 1$, and $I_0 < I_\infty$, Susceptible and infected proportions converging to equilibrium values.

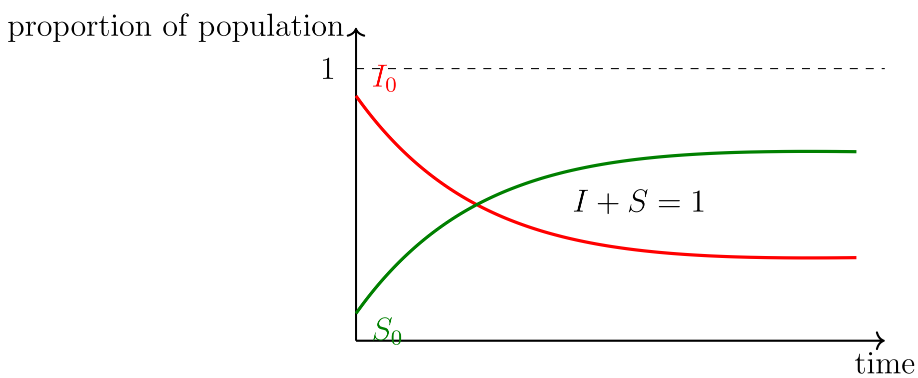


Figure 3: Visualization of Case 2 $R_0 > 1$, and $I_0 > I_\infty$ Infected proportion (red) and Susceptible proportion (green) over time for with $I + S = 1$ as the dashed line.

Summary of the two Cases

Condition	**Long term infection level**	**Qualitative behavior**
$R_0 \leq 1$	$I_\infty = 0$	infection decays monotonically to 0
$R_0 > 1$	$I_\infty = 1 - \dfrac{1}{R_0}$	infection grows or decays towards a stable infection rate

2.4 The SIR Model

The classical SIR framework partitions the population into three fractions:

$$S(t) + I(t) + R(t) = 1 \tag{10}$$

corresponding to **S**usceptible, **I**nfectious, and **R**ecovered (with permanent immunity). Movement is strictly in one direction: $S \to I \to R$.

$$\boxed{S} \xrightarrow{\quad \beta SI \quad} \boxed{I} \xrightarrow{\quad \gamma I \quad} \boxed{R}$$

Let's consider the ODE system of the SIR model:

$$\frac{dS}{dt} = -\beta SI \qquad (\leq 0) \tag{11}$$

$$\frac{dI}{dt} = \beta SI - \gamma I \tag{12}$$

$$\frac{dR}{dt} = \gamma I \qquad (\geq 0) \tag{13}$$

Where all parameters satisfy $\beta > 0,\ \gamma > 0$

Initial conditions of the model:

Typically, we would place almost everyone in S except for a small initial infected fraction in I:

$$R(0) = 0, \quad I(0) = I_0 \in (0, 1), \quad S(0) = 1 - I_0.$$

Stationary Solution to the ODE

It turns out the SIR model has two stationary solutions, the first of them being the one where $I = 0$ for all times.

a) **Trivial disease-free equilibrium**

Setting $I(t) = 0$ forces $\frac{dR}{dt} = 0$, showing that $R(t)$ must be constant. With $S+I+R = 1$ this implies that S is constant as well, and since $S_0 = 1 - I_0 = 1$, we conclude that

$$(S, I, R) = (1, 0, 0) \quad \forall t.$$

This solution exists for any choice of β, γ but only holds when the initial infection $I(0) = 0$, which is not very interesting, so let's consider the generic case $I_0 > 0$ next.

b) **Generic initial infection $I_0 > 0$**

We analyse the signs in the ODEs.

- Because $\dfrac{dS}{dt} = -\beta SI \leq 0$ and $S(t)$ remains non-negative, the susceptible curve is monotonically decreasing and converges in limit $S_\infty = \lim\limits_{t\to\infty} S(t)$.
- Because $\dfrac{dR}{dt} = \gamma I \geq 0$ and $R(t) \leq 1$, the recovered curve is monotonically increasing and converges in limit to $R_\infty = \lim_{t\to\infty} R(t)$.
- Taking limits in $S(t) + I(t) + R(t) = 1$ yields $S_\infty + I_\infty + R_\infty = 1$. If I_∞ were positive, then $\frac{dR}{dt} = \gamma I$ would stay positive, contradicting the fact that $R(t) \leq 1$ (since we cannot have a proportion greater than 1).

  Hence:

$$I_\infty = 0$$

Consequently,

$$(S, I, R) \longrightarrow (S_\infty,\, 0,\, R_\infty) \quad \text{with } S_\infty + R_\infty = 1$$

If we think about this intuitively, the infection either never starts or infects part of the susceptible pool and then dies out, leaving a permanent immune fraction R_∞ and a reduced susceptible pool S_∞.

Short–time behavior

To analyze how fast the infection grows or decays, let's consider the log-derivative of infected individuals:

$$\frac{d}{dt}\log I = \frac{1}{I}\cdot\frac{dI}{dt} = \beta S - \gamma < \beta - \gamma.$$

1. $R_0 = \frac{\beta}{\gamma} \leq 1$: Then the derivative is negative, so $I(t)$ decays exponentially.

2. $R_0 = \frac{\beta}{\gamma} > 1$: If S_0 is sufficiently close to 1, the derivative is positive at time 0, so $I(t)$ grows exponentially while S is still close to S_0. In the general case without any assumptions on S_0, the condition $R_0 > 1$ must be adapted to $R_0 S_0 > 1$.

Long–time behavior

Let's take a look back at equations (11) and (13):

$$\frac{dR}{dt} = \gamma I, \qquad \frac{dS}{dt} = -\beta SI.$$

Using $R_0 = \dfrac{\beta}{\gamma}$ we obtain

$$\frac{dS}{dR} = \frac{\frac{dS}{dt}}{\frac{dR}{dt}} = \frac{-\beta SI}{\gamma I} = -R_0\, S$$

which we can rewrite as

$$\frac{1}{S}\, dS = -\, R_0\, dR \tag{14}$$

Integrating (14) starting from the initial state $(S_0,\, R = 0)$ to the current state $(S(t),\, R(t))$ gives us:

$$\int_{S_0}^{S(t)} \frac{dS}{S} \;=\; -\, R_0 \int_0^{R(t)} dR.$$

Evaluating the integrals then gives

$$\ln\!\left(\tfrac{S(t)}{S_0}\right) \;=\; -\, R_0\, R(t),$$

i.e.,

$$S(t) = S_0\, e^{-R_0\, R(t)} \tag{15}$$

Now, we will write the equation for I in terms of R via the equation $S + I + R = 1$:

$$I(t) = 1 - S(t) - R(t) = 1 - R(t) - S_0\, e^{-R_0 R(t)} \tag{16}$$

Let's send t to the limit $t \to \infty$ to understand the long-term behavior. To this end, we note that we already showed that $I_\infty = \lim\limits_{t\to\infty} I(t) = 0$. Taking limits in (16) therefore gives the **final size equation**:

$$\boxed{1 - R_\infty = S_0\, e^{-R_0 R_\infty}} \tag{17}$$

where $S_0 = 1 - I_0$. Using the convexity of $S_0 e^{-R_0 R_\infty}$ as a function of R_∞ and the fact that for $S_0 < 1$, its intercept with the y-axis is strictly smaller than 1, one easily sees that the above implicit equation for R_∞ has exactly one solution $0 < R_\infty < 1$, with $R_\infty \searrow 0$ as $S_0 \nearrow 1$ if $R_0 \leq 1$ while it converges to a non-zero value if $R_0 > 1$, as illustrated for the green and red curves in Figure (4)

Remark 0.1. *It is not hard to show that as $S_0 \nearrow 1$, R_∞ converges to the largest solution of*

$$1 - R_\infty = e^{-R_0 R_\infty}, \tag{18}$$

which the reader will recognize as the survival probability of a Poisson(R_0) branching process, which we have shown to be zero for $R_0 \leq 1$ in our second lecture on branching processes. Writing this implicit equation in the form

$$R_0 = -\frac{1}{R_\infty}\log(1 - R_\infty) > \frac{1}{1 - R_\infty},$$

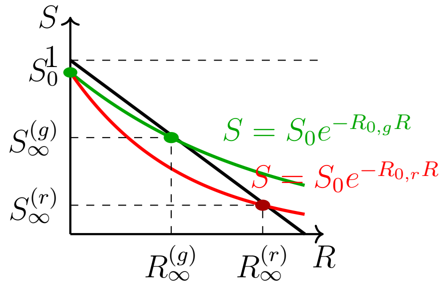


Figure 4: Final epidemic size R_∞ as $S_0 \nearrow 1$.

we furthermore see can get the lower bound $R_\infty > 1 - \frac{1}{R_0}$ for $R_0 > 1$.

Note the relationship to what is called the herd immunity threshold, *which is the point beyond which the infection starts to die out, i.e. the point where $dI/dt = 0$. From the differential equation for I, we see that this happens at the point t_0 where $\beta S(t_0) = \gamma$, i.e., $S(t_0) = \frac{1}{R_0}$. Since $R_\infty = 1 - S(\infty) > 1 - S(t_0)$, we recover the above bound.*

Summary:

- $R_0 \leq 1$: Here, the limiting final-size equation (18) reduces to the single root $R_\infty = 0$. Therefore, no large outbreak is possible and $\max_t I(t) \to 0$ as $I_0 \to 0$, with the susceptibility $S(t)$ remaining close to S_0 for all t.

- $R_0 > 1$: The limiting final-size equation (18) has two solutions, the trivial solution $R_\infty = 0$ and a positive root that satisfies $R_\infty > 1 - \frac{1}{R_0}$. The larger (non-zero) root is the survival probability of a Poisson(R_0) branching process and the limiting final size as $S_0 \nearrow 1$. Now, no matter how small I_0, the epidemic first grows, and only starts to die down once S has reached the herd immunity threshold $S(t_0) = 1 - \frac{1}{R_0}$.

We sketch the resulting dynamics for small values of $I_0 = 1 - S_0$ in Fig. 5

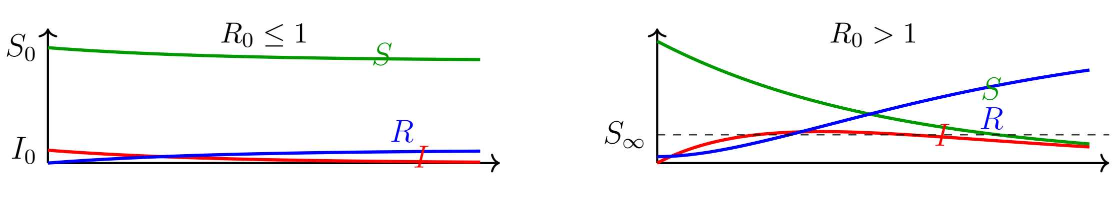


Fig. 5: The SIR epidemic curves for small, initial outbreaks $I_0 = 1 - S_0$.

3 Stochastic SIR Model

- **Population:** n labeled individuals $\{1,\ldots,n\}$, where the 3 states are Susceptible (S), Infected (I) and Recovered (R).
- **Initial condition:** Exactly one node is infected at $t=0$, let's say **individual 1** (WLOG), and all other individuals are susceptible.
- **Recovery clock:** If individual i is infected, it carries an independent exponential clock
$$T_i \sim \mathrm{Exp}(\gamma), \qquad \mathrm{P}(T_i \geq T) = e^{-\gamma T},$$
and moves to state R when its own clock rings.
- **Infection clocks:** While i is infectious it rings a rate β exponential clock for each still susceptible $j \neq i$. If the ij clock rings before i's recovery, individual j becomes infected.

The state count processes $N_S(t),\ N_I(t),\ N_R(t)$ are Continuous Time Markov Chains (CTMCs) whose Kolmogorov forward equations form a stochastic differential–equation (SDE) system. Note that analyzing the SDE is beyond the scope of our course.

3.1 Next-Generation Calculation and Forward Branching Process

Let's start by fixing an infected individual i and condition on its future recovery time $T_i = t$. While infectious, each susceptible $j \neq i$ receives infection attempts at rate β. Therefore:

$$\mathrm{P}(i \text{ infects } j \mid T_i) = \mathrm{P}(\mathrm{Exp}(\beta) \text{ clock rings before } T_i) = 1 - e^{-\beta T_i} \;=\; p_{T_i}$$

Now, let X be the number of individuals infected directly by node i. Conditional on T_i, we get:

$$X \mid T_i \;\sim\; \mathrm{Binomial}(n-1,\ p_{T_i}) \quad \text{and} \quad \mathrm{E}[X \mid T_i] \;=\; (n-1)p_{T_i}.$$

Taking expectation over the recovery time distribution we get that the expected number of people infected by an early infected individual is

$$R_0 = \;\mathrm{E}[X] = (n-1)\int_0^\infty \big(1 - e^{-\beta t}\big)\gamma e^{-\gamma t}\,dt \;=\; (n-1)\frac{\beta}{\beta+\gamma} \;\xrightarrow[n\to\infty]{}\; \frac{\tilde{\beta}}{\gamma},$$

if we scale β as $\beta = \frac{\tilde{\beta}}{n-1}$.

If we are working with a large and well mixed population, the early generations of infection can therefore be approximated by a Galton–Watson branching process with offspring mean $R_0 = \frac{\tilde{\beta}}{\gamma}$, and off-spring distribution obtained by first drawing

$$T \sim \text{Exp}(\gamma)$$

and then choosing the number of off-spring according to

$$X \sim \text{Binomial}(n-1, p_T).$$

We denote this branching process by T_{SIR} and call it the *forward branching process,* a term whose meaning will become clear in a moment.

3.2 Infection Digraph

Consider an SIR epidemic on a general graph $G = (V, E)$. When an infection spreads over such a graph, whenever a vertex i gets infected, it will start a recovery clock as well as infection clock to all its neighbors, and will pass on the infection whenever one of the infection clocks clicks before the recovery clocks. But for a mathematical analysis, nothing prevents us from pre-flipping all these exponential clocks, but only using them once the infection reaches the corresponding vertices.

This leads to the so-called infection digraph, where at time $t = 0$, we set all the exponential clocks as follows:

1. $T_i \sim \text{Exp}(\gamma)$, $\forall i \in V$
2. For each $i \in V$, set an exponential clock to all pairs $j > i$: $T_{ij} \sim \text{Exp}(\beta)$

We then define an oriented graph D_{SIR} on V by choosing

$$(i, j) \in E(D_{SIR}) \iff T_{ij} < T_i. \tag{19}$$

So, based on our definition above (19), the directed random graph D_{SIR} will contain an edge $i \to j$ if and only if i would be able to reach j, before i has recovered. Below are some observations we can make:

- Conditioned on the recovery times $T = (T_i)_{i \in V}$, edges are independent with

$$\text{P}((i, j) \in E \mid T) = p_{T_i}.$$

- If an infection starts at vertex x, then the set of infected nodes is set of all vertices that can reached from x in D_{SIR}, i.e., the out–component $C_+(x)$ of x in D_{SIR}.
- If our underlying graph is the complete graph, and we consider a BFS exploration of $C_+(x)$ we end up with T_{SIR}, at least for the first few generations, where the infection has not yet significantly reduced the number of infected vertices. Hence the name forward branching process.

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